Vadose Zone Flow and Transport in Cracked Heavy Textured Soil

By Bhaskar Joshi[1] and Charles Maulé[2]

Abstract: Investigated were physical processes governing vadose zone water flow and solute transport in a heavy-textured crack-prone soil of the semiarid Canadian Prairies. Soil moisture, soil temperature, and relevant meteorological observations were recorded over a period of about two years. Environmental chloride and tritium concentrations in the soil water were determined. Analysis of the data indicates that during snow melt, water and solutes are flushed down rapidly via cracks and fissures in the root zone. The soil at these depths is still frozen during snow melt. In late spring after the entire soil profile thaws, water and solutes move downward by a diffusion dominant advective-diffusive flow mechanism. The chloride and tritium profiles obtained within the vadose zone support this argument. This conceptual model of the flow and transport processes is supported by calculations of advective and diffusive soil water flux from chloride profiles. Simulation of the tritium profile using a simple analytical model also gives very good agreement with measured data.

INTRODUCTION

The flow of water and transport of solutes in uncracked soils may be studied by simulation models based on Darcy's law. The same is not possible for flow and transport in cracked soils (Beven and Germann 1982; Bouma and Loveday 1987). There are not many studies in these types of soils under frozen conditions, spanning episodes like snowmelt.

We recorded soil temperature, soil moisture, and environmental tracer profiles in heavy-textured cracked-soil horizons of a semiarid dryland farming area in the Canadian Prairies (Joshi 1997). Observations were made over the entire depth of the vadose zone, and spanned a period of about two years including snowmelt duration. The study was carried out in the cold continental climate of Central Canada. This paper describes the observations and the field data that were used to arrive at a conceptual model for explaining the mechanics of flow and transport.

LITERATURE REVIEW

Macropore flow in soils has been addressed widely in the literature (Thomas and Phillips 1979; Beven 1981; Bouma et al. 1981; Keller et al. 1986; Seyfried and Rao 1987; Sudicky and McLaren 1992). Macropore flow in cracked heavy-textured soils has also been studied by many workers (Bouma and Dekker 1978; Bouma et al. 1978; Hoogmoed and Bouma 1980; Singh and Kanwar 1991). White (1985) suggested that macropores can result in rapid transport of solutes to substantial depths below the ground. It has been reported that in deep vadose zones, which are common in arid and semiarid regions, preferential flow may be important near the ground surface (Gee and Hillel 1988). At greater depths, the flow and transport may be slow and diffusive (Barnes et al. 1994). During episodes like thunderstorms and snow melt, rapid movement of water and solutes occurs through the cracked surficial horizons, after which the much slower movement may occur (Stephens 1994). Similar studies of flow and transport in frozen soils under cold climatic conditions are few.

Many studies have reported the interaction of water and solute movement in freezing soils under closed-system laboratory conditions (Hoekstra 1966; Dirksen and Miller 1966; Cary and Mayland 1972). There are relatively few reports on moisture flow and solute transport in frozen soils under field conditions (Cary et al. 1979; Gray et al. 1985; Gray and Granger 1986). Gray and Granger (1986) studied moisture and solute movement during winter in fallow and stubble fields near Outlook and Saskatoon in Saskatchewan, Central Canada. They reported upward capillary movement of water into the frozen zone of about 0.8 m depth until a pore saturation of 80–90% was reached. Their study was carried out in uniform silty-loam and silty-clay soils. The water table was located at depths of 2.5–4 m, and conditions were conducive to capillary movement (Gray and Granger 1986). Their study did not span the snowmelt period, and changes below the root zone were not recorded.

SITE DESCRIPTION

The experimental area of the study reported here is located in the central part of the province of Saskatchewan between 52 and 53° North Latitude and 106 and 108° West Longitude (Fig. 1). The province of Saskatchewan forms part of the physiographic area known as the Great Plains Province of the Interior Plains of North America.

The study was carried out on an instrumented plot at the University of Saskatchewan's Kernen Farm located about 12 km east of Saskatoon, Canada (Fig. 1). Saskatoon has a continental climate characterized by a wide range in temperatures between summer and winter as well as by short-term changes in the weather. The mean annual precipitation is about 371 mm and the lake evaporation about 718 mm (Wittrock et al. 1989). Roughly 30% of the annual precipitation occurs as snow. The mean monthly temperature ranges from −19.1°C in January to +18.4°C in July.

Regional Geology

The shallow stratigraphic section obtained from test hole data around the study area (within a radius of 3 km) indicates that the land surface is underlain by 5–6 m of glacio-lacustrine clay and clay loam which is underlain by till to a depth of about 18 m (Christiansen 1970). The Forestry Farm aquifer is encountered at around this depth, and it is underlain by till material.

The tills under the glacio-lacustrine layer are of Pleistocene age (Keller et al. 1986). The till member underlying the glacio-lacustrine material has been classified as the Floral Formation of the Saskatoon Group (Christiansen 1970). The texture of the Floral formation ranges from sandy loam to clay loam.

[1]Sr. Hydrogeologist, Water Management Consultants Inc., 333 N. Wilmot Rd., Ste. 310, Tucson, AZ 85711. E-mail: kissh@gci-net.com

[2]Assoc. Prof., Dept. of Agr. and Bioresour. Engrg., Univ. of Saskatchewan, Saskatoon, Canada S7N 1L5.

Note. Discussion open until November 1, 1999. To extend the closing date one month, a written request must be filed with the ASCE Manager of Journals. The manuscript for this paper was submitted for review and possible publication on January 6, 1998. This paper is part of the ***Journal of Irrigation and Drainage Engineering***, Vol. 125, No. 3, May/June, 1999. ©ASCE, ISSN 0733-9437/99/0003-0122–0130/$8.00 + $.50 per page. Paper No. 17328.

J. Irrig. Drain Eng. 1999.125:122-130.

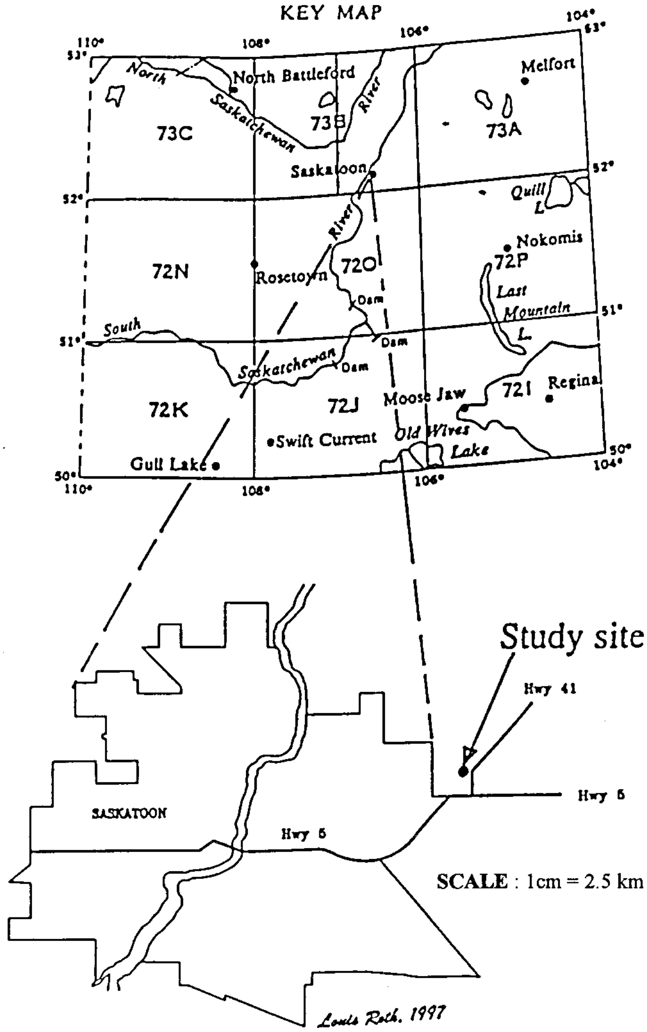

FIG. 1. Topographic Location of Site

The sand content ranges from 40–45% while silt content ranges from 30–40% (Christiansen 1970). The upper part of this formation (upper till) is hard and heavily stained with iron and manganese oxides on joint surfaces, although there is no evidence of leaching (Christiansen 1970). The Forestry Farm aquifer is an intratill aquifer located between the upper and lower Floral tills. This aquifer was encountered at the lower slope position of the study site. A Saskatchewan Research Council test hole section located about 2.25 km north of the site (SRC No. NW-16-08-37-4W3) indicates that the upper till is friable and fracture surfaces are stained with iron oxide.

REGIONAL HYDROGEOLOGY

A range of low hills called Strawberry Hills that run along a north-south axis roughly 5 km to the east of the study site serve as a major recharge area (Christiansen 1970). The Strawberry Hills area provides direct recharge into the Forestry Farm aquifer. The aquifer extends out in the shape of an elongated disc from the Strawberry Hills to eastern bank of the South Saskatchewan River which flows in a rough north-south direction (Christiansen 1970). It pinches out in the vicinity of Kernen farm, which lies roughly toward the southeast. The zone above the Forestry Farm aquifer is a recharge zone as there is no indication of artesian conditions. The well-drained and productive soils at Kernen farm are also indicative of recharge (Christiansen 1970).

The surficial stratified sand, silt, and clay deposits that overlie the uppermost till present have been termed ‘‘surficial aquifers’’ by Christiansen (1970). Surficial aquifers were deposited during or subsequent to, the final deglaciation of the Saskatoon area (about 10,000 years ago). Water in these aquifers originates as precipitation that has infiltrated downward from the ground surface to the water table, which forms the upper surface of the surficial aquifers (Christiansen 1970). Recharge occurs seasonally, mainly during snowmelt period or prolonged rainstorms. The water then moves downward and laterally to recharge deeper aquifers or discharges to local topographic depressions (Christiansen 1970). The water level encountered in piezometers installed at the study site is within the surficial clay and silt deposit except in one deep piezometer (at 20 m depth), which probably intercepts the Forestry Farm aquifer.

MATERIALS AND METHOD

Fig. 2 shows the instrumentation installed at the study site. Five tensiometers at depths ranging from 0.7–6 m below ground level were installed at the upper, middle, and lower slope locations. Soil temperature probes were installed at depths ranging from 0.1 to 3.5 m at the upper and lower slope locations. Piezometers were installed at all the slope locations. In addition, three neutron probe access tubes to a depth of about 3 m were installed at each slope position. Precipitation data were collected at the site, and other climatic data were recorded at a weather station located about 1 km away from the site.

Chemical analyses were carried out by the senior writer on core samples collected during installation of the piezometers. Tritium analyses were carried out at the Environmental Isotope Laboratory, University of Waterloo, Ontario, Canada. Soil Water chloride concentrations were measured on saturated paste extracts using ion selective electrodes (Carter 1993). The expected accuracy of ion selective electrodes is 0.1 mV with a decadal concentration change of 59 mV, for univalent anions (Camann 1979). Tritium was directly measured by liquid scintillation counting on the extracted soil water samples (Drimmie et al. 1993). These measurements have a lower limit of detection of approximately 8 tritium units.

OBSERVATIONS

The experimental plot of about 3 ha is flat and gently inclined downward from east to west with a slope of about 2% (Fig. 2). In further discussion, the eastern end of the plot is referred to as the *upper slope* position, and the western end, near the bottom of the slope, is called the *lower slope* position. The middle slope position is roughly midway between the upper and lower slope locations. The bottom of the slope (about

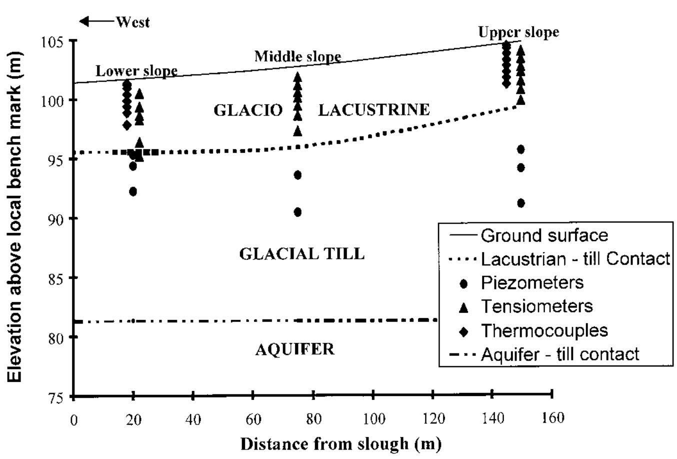

FIG. 2. Shallow Stratigraphic Section and Site Instrumentation (Local Bench Mark = 100.00 m Was Concrete Culvert about 100 m South of Lower Slope)

J. Irrig. Drain Eng. 1999.125:122-130.

30 m away from the lower slope position) is a local depression in the landscape where water accumulates after snowmelt.

The Kernen farm landscape is characteristically level to undulating glacio-lacustrine plain with minor areas of hummocky topography. The farm is situated in the Dark Brown soil zone (Souster 1979). Salinity is not a serious problem, though moderately high salt levels may occur in silty lacustrine materials (Souster 1979). Figs. 3(a and b) show the depth profile of sand and clay percentages at the various slope locations. The clay percentage peaks at 1 m, 3 m, and 3 m depth at the lower, middle, and upper slopes. The sand content appears to increase

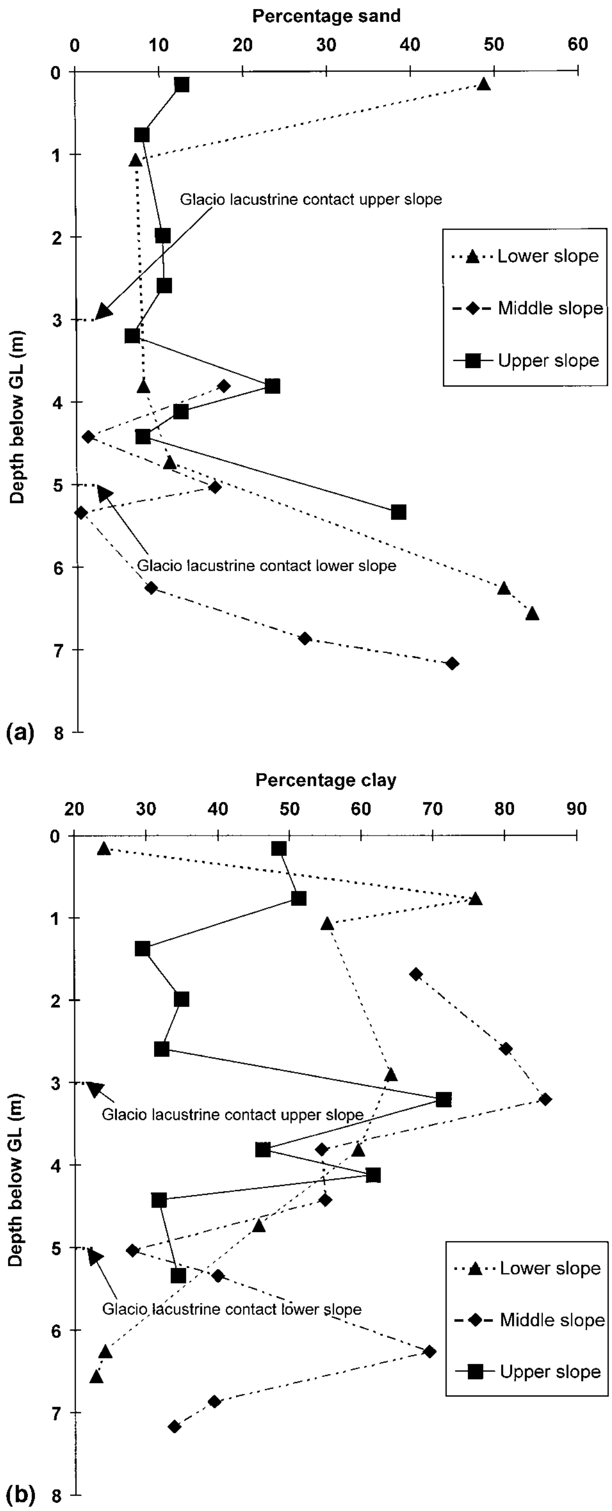

FIG. 3. (a) Variation of Sand Content with Depth at Kernen; (b) Variation of Clay Content with Depth at Kernen

with depth at all the slope positions after the glacio-lacustrine contact is encountered.

The upper 3–6 m of the glacial deposit at the study site are highly weathered. The depth of the weathered zone is indicated by a color transition in the clay matrix from brown to grey at around this depth. The ''A'' horizon is gleyed at the lower slope position. The effects of weathering extend down to a depth of about 5.5–6 m at upper and middle slope positions and to about 3 m at the lower slope position.

Isolated root hairs were observed within the soil cores to a depth of about 4–5 m. Iron oxide stains were observed to about 7–11 m depth. The influence of these features on flow and transport mechanisms has to be viewed in the light of additional evidence, for example from solute profiles (Keller et al. 1986). Some of this evidence is briefly discussed in the following paragraphs.

In a study carried out in the same till unit about 10–30 km away from Kernen farm, Keller et al. (1986) concluded that fractures were hydraulically active. However, the smooth solute profile (tritium in this case) indicated that the till behaved like an equivalent porous medium because of matrix diffusion between finely spaced fractures (Keller et al. 1986). The presence of fractures and hydraulically active till units would normally imply highly asymmetric and multipeaked solute profiles (Sudicky 1983; Hendry 1988), which were not observed at their site. Thus the influence of fractures on flow and transport mechanisms was different than what one might deduce based on the presence of fractures and the hydraulic conductivity data. The increase of sand content with depth at Kernen [Fig. 3(a)] is also significant in this context. Li and Ghodrati (1997) have suggested that preferential flow tends to occur to a greater extent in fine-textured soils than in coarser-textured soils probably because of the finer soil's lower matrix hydraulic conductivity.

At a farm located about 5 km away from Kernen, Dasog (1986) determined that the average depth of cracks was 47 ± 30 cm in a similar soil horizon (cropped to grass). Other workers (specifically, the co-author) have observed deeper cracks up to 1 m depth beyond the lower slope position at Kernen. Thus seasonal desiccation cracks at Kernen may not extend much below the root zone (about 1.2 m depth), while freeze thaw cracks extend to about 1.5 m since the zero degree isotherm penetrates to roughly this depth during the cold season. Soil core samples obtained at the site indicate that weathering features in the form of finely spaced microcracks and iron oxide staining extend down to about 11 m depth.

Available historical data on land use at Kernen farm show that land was broken for cultivation about 100 years ago. The farm was handed over to the University of Saskatchewan in 1977 and no records of land management are available prior to this. The available records show that the area of the experimental plots remained fallow during 1977–1979 and in 1981. It has been cropped in all other years. Dryland farming is practiced in the study area and crops are not irrigated.

Figs. 4(a–d) show the variation of soil temperature and volumetric water contents with depth over the duration of this study. Figs. 5 and 6 show the temporal variation of water content with time at depths of 0.6 m and 0.9 m below ground level. A comparison of soil core and neutron probe volumetric water contents on nearby dates is given in Table 1. It is clear that at depths below 0.5 m these two determinations of volumetric water content fall within 10% of each other. Thus, neutron probe readings may be considered fairly reliable. However, they represent average values within a sphere of radius 15–30 cm. The higher the water content the smaller is the radius sampled. The percentage saturation below 1.5 m depth was above 50% at the upper slope and above 93% at the lower

J. Irrig. Drain Eng. 1999.125:122-130.

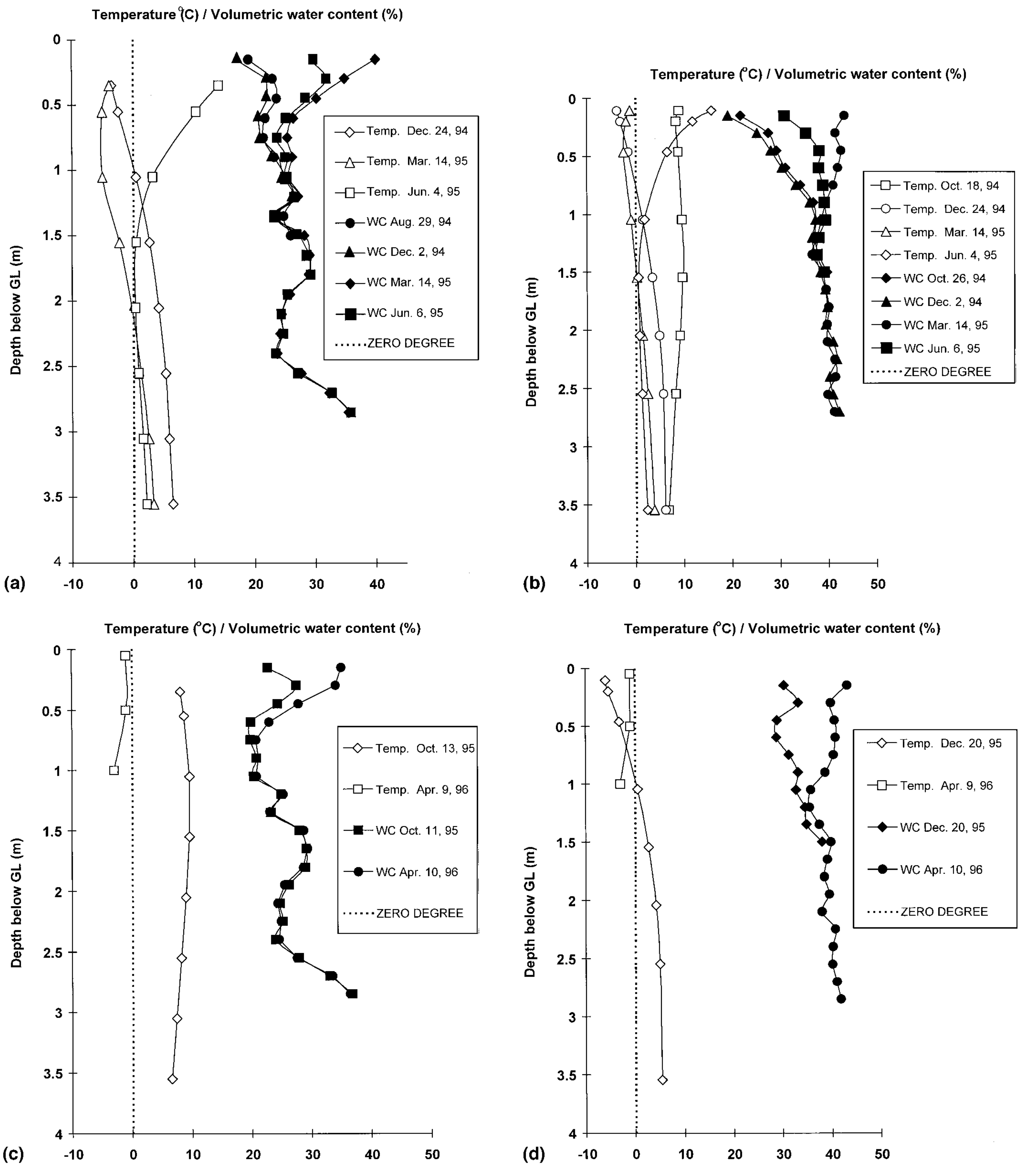

FIG. 4. Observed Soil Temperature and Volumetric Water Content Profiles at (a) Upper Slope 1994–95; (b) Lower Slope 1994–95; (c) Upper Slope 1995–96; (d) Lower Slope 1995–96

slope. Thus, for the relatively high water contents at depths below 1.5 m, the sampling radius may be nearer to 15 cm.

The chloride profiles at the upper middle and lower slope positions are shown in Fig. 7. These profiles show sharp, well-defined bulges well above the water table beginning at depths ranging from 1.8 m at the upper and middle slopes to 2.4 m at the lower slope position.

This chloride accumulation is probably the result of accession from precipitation over several thousand years (Scanlon 1991; Joshi 1997). It was calculated that a period of 4,000 to 7,000 years would be required for accumulation of the chloride observed in the profiles (Joshi 1997; Scanlon 1991).The sediments in the study area are glacio-lacustrine and do not contain any chloride of marine origin (Christiansen 1970). The low observed chloride concentration in soil water also indicates that chloride is not in equilibrium with chloride minerals, and any in situ source may be negligible (Scanlon 1991). Assuming that there was in fact some chloride in sediments at

J. Irrig. Drain Eng. 1999.125:122-130.

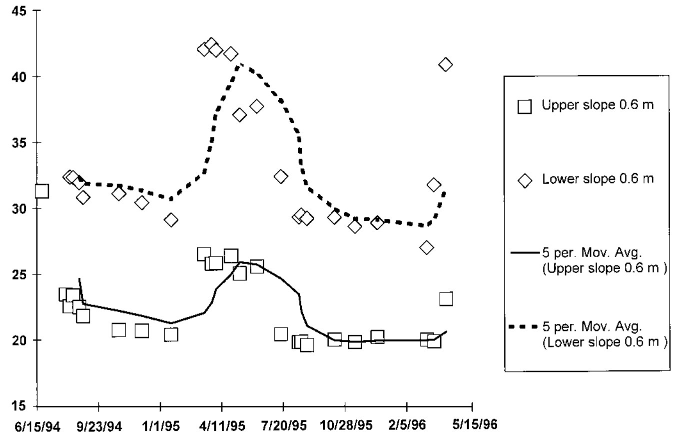

FIG. 5. Temporal Variation of Water Content at 0.6 m Depth during 1994–95

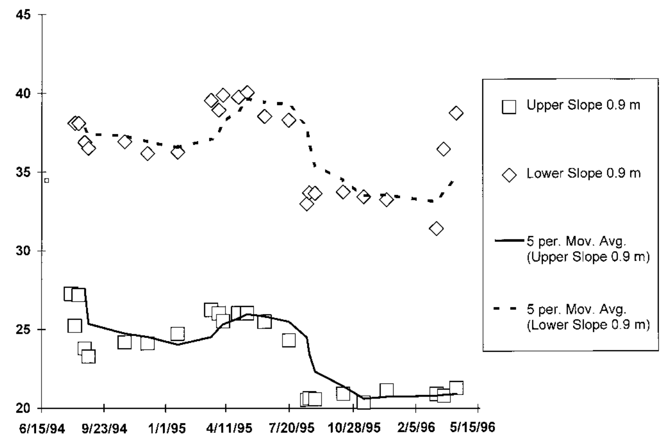

FIG. 6. Temporal Variation of Water Content at 0.9 m Depth during 1994–95

TABLE 1. Comparison of Core and Neutron Probe Volumetric Water Contents (θ) at Upper Slope Position on Nearby Dates

Depth (m) (1)	Core sample on July 27, 1994 θ (2)	Depth (m) (3)	Neutron probe on August 1, 1994 θ (4)
0.46	0.22	0.45	0.24
0.76	0.26	0.75	0.24
1.06	0.31	1.05	0.29
1.37	0.33	1.35	0.24
1.68	0.28	1.65	0.30
2.28	0.16	2.25	0.22

the time of deposition, it is possible that the chloride bulges formed as a result of redistribution of an original uniformly low chloride concentration in the profile.

A rough calculation assuming that initially there existed a uniform chloride concentration equal to what now prevails below the bulge, shows that it would be possible to accommodate only about 25% of the cumulative chloride available in the profile, within the zone currently occupied by the bulges. For example, the total amount of chloride in the upper slope profile may be calculated by assuming a uniform chloride concentration of 25 mg/L below the chloride bulge (Fig. 7) and an average volumetric water content of 34% at this depth (obtained from core samples). This gives a total chloride content of about 45.3 mg. The total amount of chloride in the observed chloride bulge (Fig. 7) may be calculated from the observed volumetric water contents and the chloride profile. This gives a value of about 261 mg.

Fig. 8 shows the tritium profile obtained at the lower slope

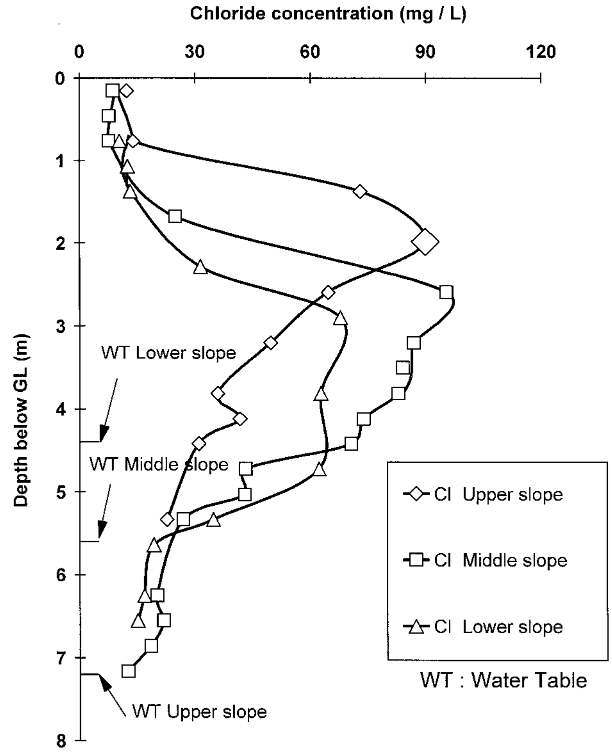

FIG. 7. Chloride Concentrations in Soil-Water

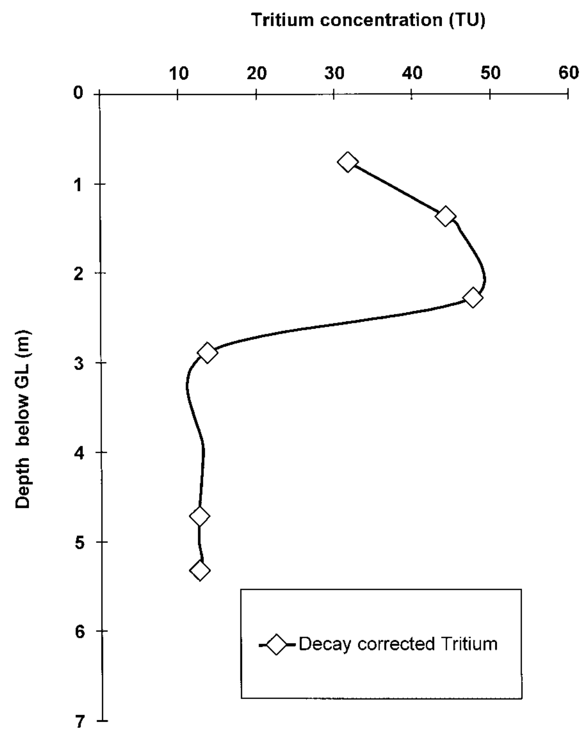

FIG. 8. Profile of Tritium Concentration in Soil-Water at Lower Slope and Corresponding Volumetric Water Content

position. A clear and prominent tritium peak at a depth of about 2.2 m was revealed by the data. Decay correction in this profile was made to allow for the time elapsed between the sampling and analysis of water samples. The tritium half life was taken as 12.3 years. Relatively high tritium levels below the peak (11 TU) down to a depth of 5 m indicate that the glacio-lacustrine zone may contain recent (post-1953) waters, although this value is close to the cut-off value of about 9 TU for pre-1953 water (Hendry 1988).

Tritium concentrations in precipitation have been influenced

J. Irrig. Drain Eng. 1999.125:122-130.

by nuclear testing. Pre-1953 tritium concentration in precipitation was 5–10 TU (Payne 1972). The concentrations increased dramatically during 1953 to 1969 due to thermonuclear testing. Maximum values of more than 3,000 TU were recorded in monthly composite samples in the northern hemisphere (Solomon et al. 1995). Allowing only for radioactive decay, any waters which entered the subsurface before 1953 would have a maximum tritium concentration of about 1 TU at the present time (tritium decay constant of 0.0565 per year; half life 12.43 years). Waters that entered the subsurface after 1953 will have higher tritium contents. Since the analytical precision of tritium measurements is ±8 TU, water samples with concentrations of greater than 9 TU are considered as post-1953 water.

The lower tritium levels nearer to the ground surface (Fig. 8) are indicative of leaching by more recent water having low concentrations of tritium. Current levels of tritium in precipitation vary from less than 1, to 3 TU (International Atomic Energy Agency 1991).

DISCUSSION

It is evident from Figs. 4(a–d) that the soil was still frozen to a depth of about 1–1.5 m (around middle March to early April) when a dramatic increase in water content occurred due to snow melt. In 1995, air temperatures went above zero in early March, and sustained snowmelt had started after April 11. An early melt (above zero air temperatures for more than five days) had occurred in early March. In 1996, sustained snowmelt started on April 4 but an early melt had occurred February 5–15. The phenomenon of abrupt increase in water content within the root zone was observed in 1995 as well as 1996. It may be inferred that melt water infiltrated to at least 1.5 m depth during snow melt episodes.

A close look at the melt (March) water content profiles during 1994–1995 [Figs. 4(a,b)] shows an accumulation of 23 and 48 mm moisture within the top 1.5 m depth since that of fall (October), at the upper and lower slope positions, respectively. During the period of October 26, 1994, to March 14, 1995, the total precipitation was 48 mm. Similarly during 1995–96 there was an accumulation of 31 mm moisture within the top 1.5 m depth of the soil profile from October to April at the upper slope and 88 mm moisture (within the 1.5 m depth interval) from December to April at the lower slope position [Figs. 4(c,d)]. During the period October 11, 1995, to April 10, 1996, the total precipitation was 86 mm. On the basis of these observations one may say that in these two years, during melt, almost all of the over winter precipitation ended up in the soil at the lower slope position while nearly 35–50% infiltrated at the upper slope. The estimated infiltration at the lower slope position may actually be lower, since runoff may account for some of the increase.

However, very little of this large pulse of melt water appeared to penetrate downward below the 1.5 m depth [see Figs. 4(a,b), June profile]. Note that the soil profiles were still frozen to about 1.5 m depth until March, when first melt occurred. The entire frozen profile did not reach above zero temperatures until early June. By that time the water content profiles had already receded considerably [Figs. 4(a,b)]. It is reasonable to expect that most of the melt water pulse either traveled down the slope as runoff and interflow above the 1.5 m depth or was lost as evapo-transpiration during this period.

One may speculate that moisture and solutes may move downward without necessarily changing the soil moisture content, or the changes may be beyond the neutron probe resolution. However, the nature of solute profiles (Figs. 7 and 8) does not support the possibility of short-term steady-state soil water movement at these depths where negligible changes in soil moisture content are observed. In a semiarid region, water flux is expected to decrease within the root zone due to evaporation and extraction by plant roots. After a certain depth when evapo-transpiration effects become negligible, water flux may become nearly constant. It may be shown that under steady-state water and solute flux with advection as the primary mode of transport, one would expect a monotonously increasing solute content that should become nearly constant after a certain depth (Gardner 1967). Thus the presence of chloride bulges does not support the hypothesis of steady-state flow.

In brief, the above data show clear evidence for rapid transport of the melt water pulse to depths of about 1.5 m. Figs. 5 and 6 show that the water moved rapidly to depths of up to 0.9 m within a very short period of time. The rate of increase was more pronounced at the upper slope location. Relatively slower advective-diffusive transport is hypothesized to occur below that depth. The justification for the advective-diffusive transport at depth greater than about 1.5 m may be substantiated on the basis of observed solute profiles (Figs. 5 and 6). Calculation of advective flux and associated Peclet numbers from the chloride profiles indicated that the advective-diffusive mechanism is operative. This is explained below.

Chloride profiles (Fig. 7) were used to estimate the steady-state moisture flux and Peclet numbers. A variation of the chloride mass balance method was used for these calculations (Gardner 1967; Peck et al. 1981; Scanlon 1991). Details regarding these calculation procedures are available elsewhere (Peck et al. 1981; Scanlon 1991), and only a brief description is given below.

The steady state advective-dispersive solute flux may be expressed as

$$J_s = -D_s \frac{\partial c}{\partial z} + cq_w \tag{1}$$

where J_s = mass flux density of solute ($ML^{-2}T^{-1}$); D_s (L^2T^{-1}) = hydrodynamic dispersion coefficient of solute in soil; c = solute concentration in soil water (ML^{-3}); q_w = Darcian soil-water flux (LT^{-1}); and z = elevation.

Eq. (1) may be rearranged to express directly the soil-water flux as

$$q_w = \frac{1}{c}\left[J_s - D_s \frac{\partial c}{\partial z}\right] \tag{2}$$

In (2), the first term on the right-hand side (after opening the brackets) denotes the advective moisture flux and the second term represents the dispersive flux. The hydrodynamic dispersion coefficient ($D_s = D_m + D_e$) is composed of the mechanical dispersion coefficient (D_m) and the effective molecular diffusion coefficient (D_e). Mechanical dispersion reflects the influence of local velocity variations and soil heterogeneity. Normally D_m is negligible for flow velocities less than about 0.7 my^{-1} (Scanlon 1991). The coefficient D_e includes the effects of tortuosity and water content (Porter et al. 1960). For silts, sands, and gravel it is primarily dependent on water content (Scanlon 1991). For chloride diffusion in clay soils, empirical relationships for D_e fitted to experimental data are available (Peck et al. 1981).

If a steady-state situation is assumed with a constant solute flux equal to the rate of accession from precipitation, (2) may be used to obtain q_w at any depth provided that the solute accession rate (J_s) and the solute concentration in soil water (c) are known. The total water flux (q_w) may be split into advective and dispersive soil water flux using (2). The relative importance of advective to dispersive flux can then be established by calculating the Peclet number from the estimated flux values. Peclet number characterizes the magnitude of advective relative to diffusive transport. It may be calculated as Pe = V^2T/D_s where V denotes the pore water velocity, T is the

J. Irrig. Drain Eng. 1999.125:122-130.

time span involved (maximum 10,000 years in this case), and D_s is the hydrodynamic dispersion coefficient. Advective transport is negligible if Pe << 1.

The depth profile of soil-water flux was calculated by using (1) and (2) (Fig. 9), and the depth profile of time required for chloride accumulation (Scanlon 1991; Joshi 1997) was utilized to calculate the Peclet number. Fig. 9 shows that advective soil-water flux was relatively high in the surficial zone up to about 0.5 m depth and decreased to a minimum between about 2.5–4 m depth. Thereafter it increased with depth indicating increased advective contribution arising as a result of higher hydraulic conductivity of the till or direct injection via cracks and fissures. The soil texture analyses show an increase in sand content after the glacio-lacustrine to till contact is reached at a depth of about 3 m at the upper slope and around 5 m at the lower slope [Fig. 3(a)].

The variation of Peclet number with depth is presented in Table 2. The calculated Peclet numbers (Table 2) are also indicative of diffusion dominant transport, especially below the 2 m depth (Pe << 1). An increase in Peclet numbers below 4 m depth is indicative of higher hydraulic conductivity or preferential flow contributions.

The smooth and nearly symmetric solute profiles indicate that they are a result of a flow process where diffusion dominates and flow velocities are not high (Joshi 1997). Decreasing clay content at depth beyond about 2–3 m [Fig. 3(b)] suggests that preferential flow may become less important with depth (Li and Ghodrati 1997).

The calculated downward advective soil-water flux ranged from about 1×10^{-11} to 1×10^{-12} m s^{-1} while diffusive soil-water flux varied from $+1 \times 10^{-12}$ to -1×10^{-12} m s^{-1}, suggesting that, in general, advective and diffusive water fluxes may be of comparable magnitude. Hence, in view of the observed solute profiles and flux calculations it may be reasonably expected that the flow and transport process below the root zone is advective-diffusive with diffusion dominant solute transport.

Tritium profiles at the upper and lower slopes show a prominent and smooth peak at a depth of 2.2 m below ground (Fig. 8). A heat flow analogy (Carslaw and Jaeger 1969, p. 62, equation 15) was used to simulate these profiles. This heat flow solution is for conduction of a slug of heat energy into a medium of thermal diffusivity k which is initially at zero temperature. Specifically it is stated that region $a < z < b$ is initially at a constant temperature V (being analogous to initial solute concentration), and the regions $0 < z < a$ and $z > b$ are at 0°C, while the surface $z = 0$ is maintained at zero for time $t > 0$. The analytical solution for transient temperature distribution (T) is given as

$$T = \frac{V}{2}\left\{\operatorname{erf}\frac{(z-a)}{2\sqrt{kt}} + \operatorname{erf}\frac{(z+a)}{2\sqrt{kt}} - \operatorname{erf}\frac{(z-b)}{2\sqrt{kt}} - \operatorname{erf}\frac{(z+b)}{2\sqrt{kt}}\right\} \quad (3)$$

The simulation was carried out by assuming that each year a 4 cm slug of water with a tritium concentration equal to the yearly average of precipitation at Ottawa (International Atomic Energy Agency 1991) moves rapidly down to a depth of about 150 cm, and after this travels by matrix diffusion. This has happened every year beginning from 1963 until 1994 and the effect of consecutive years' tritium input is added.

It was assumed that rapid movement of 98% of the recorded tritium in precipitation to a depth of 1.65 m occurs first, followed by diffusion beginning at that depth. This could explain the observed profile (Fig. 10). In order to explain the uniform tritium concentration below about 3 m (Fig. 8), 2% of the recorded tritium was considered as directly injected at depths below 3 m. The tritium input function is considered reliable. Therefore this calculation strongly supports the hypothesis of rapid water and solute movement through macropores in the root zone and slower advective-diffusive transport below that depth.

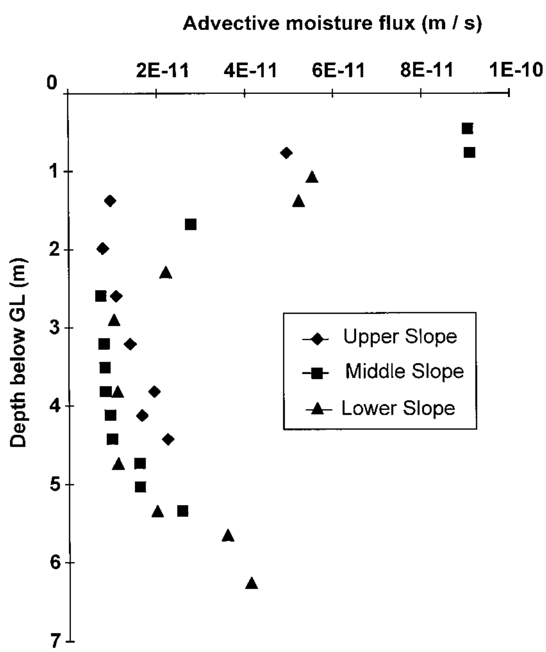

FIG. 9. Steady-State Advective Moisture Fluxes

TABLE 2. Variation of Peclet Numbers with Depth

Average depth (m) (1)	Pe (2)	Location (3)
0.5	0.19	Middle slope
3.0	0.01	Upper, middle, lower slope
4.0	0.30	Middle slope

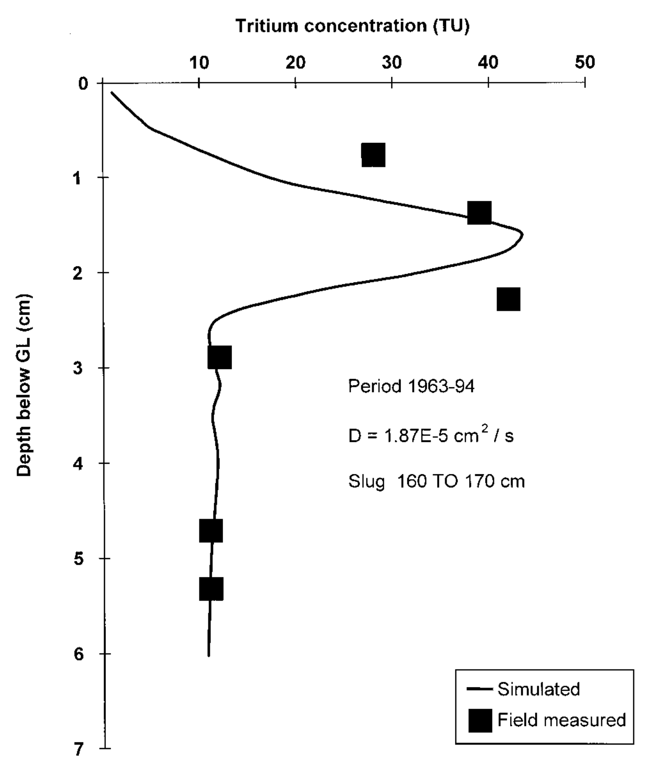

FIG. 10. Simulated and Observed Tritium Profiles at Lower Slope Position

J. Irrig. Drain Eng. 1999.125:122-130.

Conceptual Model of Vadose Zone Flow and Transport

As stated earlier, about 30% of the annual precipitation in the study area occurs as snow. Hobbs and Krogman (1971) reported that about 30% of the over winter precipitation (from harvest to seeding) on nonirrigated land was stored within the root zone. Their study was carried out on Brown Chernozem farmland in the neighboring state of Alberta. At a study site with similar soil conditions located about 600 km North of Saskatoon, Maulé et al. (1994) determined that in eluviated Black Chernozem dryland farming area, about 27% of the root zone soil moisture was made up of snow-water. Thus there is ample evidence indicating significant infiltration of over winter precipitation into the root zone.

During snowmelt the ground becomes very soft, as the upper 10 to 20 cm thick soil layer in the ''A'' horizon warms up. However, the subsoil (below the A horizon) is still frozen and its matrix hydraulic conductivity remains low. It is well known that the hydraulic conductivity of fine-grained frozen soil is in the range of 1×10^{-12} to 1×10^{-13} ms^{-1} (Cary and Mayland 1972; Oliphant et al. 1983) at temperatures near $-0.3°C$ to $-2.0°C$. Therefore a dramatic increase in water content immediately after snowmelt up to a depth of 1.5 m may not occur in uncracked frozen soils. The extent of downward water movement in an uncracked frozen soil profile should be small. As a result, large quantities of snowmelt water will runoff, causing some erosion (Chanasyk and Woytowich 1987; Maulé et al. 1993), and significant ponding in numerous small depressions typical of the glaciated landscape.

Heavy-textured soils generally crack as a result of drying toward the end of the crop season. If the rainfall in early fall is also low, these cracks may not seal and remain open into the next spring season until snowmelt. The snowmelt water can rapidly move downward as well as laterally through these cracks. Recent work (Rhoades et al. 1997; Shouse et al. 1997) has shown that during irrigation, solutes may be transported down and then horizontally within the root zone via an extensive network of cracks and fractures that form in heavy textured cracking soils. The study of Rhoades et al. (1997) was carried out in the irrigated farmland of Imperial Valley, California.

Our study was carried out in the dryland farming area of the Canadian Prairies where snow melt may be considered as an episode similar to irrigation. Observations during snowmelt at Kernen showed that the cracked soil profile behaves in a manner similar to a cracked heavy-textured soil profile in irrigated farmland areas (Rhoades et al. 1997). The extent of lateral flow was not verified during our study, although it is certainly possible.

We contend that transmission of snowmelt water occurs down the cracks that may form at the end of the previous cropping season. At depths of about 1.5 m, where these cracks close, melt water may again freeze since soil at these depths is frozen until early summer. At Kernen the soil below the root zone is not heavily cracked but contains finely spaced closed fissures that may be a result of freeze thaw cycles (Joshi 1997). These horizons approach the behavior of an equivalent porous medium and their bulk hydraulic conductivity is low (also see Keller et al. 1986). As the entire soil profile thaws progressively in late spring, the solutes swept down with melt water move by an advective-diffusive transport mechanism. The solute profiles indicate that diffusion may be the dominant transport mechanism below the root zone.

CONCLUSIONS

The soil cores obtained at our study site showed clear evidence of cracks, root holes, and freeze thaw features. The field observations on water contents and solute profiles suggested that during episodes like snowmelt, movement of water and solutes occurred rapidly to about 1.5 m depth. Computation of soil-water flux from chloride profiles showed that (1) soil-water movement was primarily advective-diffusive within the vadose zone, and (2) advective and diffusive fluxes were of comparable magnitude. This, combined with the observation that solute profiles were nearly symmetric, suggested that diffusion may be the dominant transport mechanism below the root zone. Calculation of Peclet numbers from chloride supported these conclusions. The observed tritium profile was simulated by assuming that rapid injection of water and solutes occurred to a depth of about 1.65 m, followed by diffusion dominant transport. A simple heat flow analogy was used and the simulated profile matched the observed profile closely.

Flow and transport in heavily cracked clay soils at our study site was explained satisfactorily by using a conceptual model proposed from analysis of field observations. Thus, conceptual modeling of a study area is important before choosing the analytical/numerical model. A careful record of field observations on soil cores was instrumental in establishing the conceptual model. In our case, relatively simple analytical models like the chloride mass balance equations and a heat flow analogy were sufficient to explain the observed solute profiles. Earlier studies on the flow and transport behavior of similar soils also provided guidance and support for the conceptual modeling effort.

APPENDIX. REFERENCES

Barnes, C. J., Jacobson, G., and Smith, G. D. (1994). ''The distributed recharge mechanism in the Australian arid zone.'' *Soil Sci. Soc. of Am. J.*, 58(1), 31–40.

Beven, K. (1981). ''Micro, meso, macroporosity and channeling flow phenomena in soils.'' *Soil Sci. Soc. of Am. J.*, 45, 1245.

Beven, K., and Germann, P. (1982). ''Macropores and water flow in soils.'' *Water Resour. Res.*, 18(5), 1311–1325.

Bouma, J., and Dekker, L. W. (1978). ''A case study on infiltration into dry clay soil I: Morphological observations.'' *Geoderma*, 20, 27–40.

Bouma, J., Dekker, L. W., and Wosten, J. H. M. (1978). ''A case study on infiltration into dry clay soil II: Physical measurement.'' *Geoderma*, 20, 41–51.

Bouma, J., Dekker, L. W., and Muilwijk, C. J. (1981). ''A field method for measuring short circuiting in clay soils.'' *J. Hydro.*, 52, 347–354.

Bouma, J., and Loveday, J. (1987). ''Characterizing soil water regimes in swelling clay soils, Vertisols: Their distribution, properties, classification and management.'' Texas A&M Univ. Press, College Station, Tex., 83–96.

Camann, K. (1979). *Working with ion selective electrodes-chemical laboratory practice*. Springer, N.Y.

Carslaw, H. S., and Jaeger, J. C. (1969). *Conduction of heat in solids*. Clarendon Press, Oxford, U.K.

Carter, M. R., ed. (1993). ''Soil sampling and methods of analysis.'' Lewis, Boca Raton, Fla.

Cary, J. W., and Mayland, H. F. (1972). ''Salt and water movement in unsaturated frozen soil.'' *Soil Sci. Soc. of Am. Proc.*, 36, 549–555.

Cary, J. W., Papendick, R. I., and Campbell, G. S. (1979). ''Water and salt movement in unfrozen soils: Principles and field observations.'' *Soil Sci. Soc. of Am. J.*, 43, 3–8.

Chanasyk, D. S., and Woytowich, C. P. (1987). ''Sediment yield as a result of snowmelt runoff in the Peace River Region.'' *Can. Agric. Engrg.*, 28, 7–13.

Christiansen, E. A., ed. (1970). ''Physical environment of Saskatoon.'' *Publ. No. 11378*, National Research Council of Canada, Ottawa, Ont., Canada.

Dasog, G. (1986). ''Properties, genesis and classification of clay soils in Saskatchewan,'' PhD thesis, Dept. of Soil Science, University of Saskatchewan, Saskatoon, Canada.

D'Astous, A. Y., Ruland, W. W., Bruce, J. R. G., Cherry, J. A., and Gilham, R. W. (1989). ''Fracture effects in the shallow groundwater zone in weathered Sarnia-area clay.'' *Can. Geotech. J.*, 26 43–56.

Dirksen, C., and Miller, R. D. (1966). ''Closed system freezing of unsaturated soil.'' *Soil Sci. Soc. of Am. Proc.*, 30, 168–172.

Drimmie, R. J., Heemskerk, A. R., and Johnson, J. C. (1993). ''Tritium

J. Irrig. Drain Eng. 1999.125:122-130.

analysis. Technical procedure 1.0, Rev 03.'' Dept. of Earth Sciences, University of Waterloo, Waterloo, Ont., Canada.
Gardner, W. R. (1967). ''Water uptake and salt distribution patterns in saline soils.'' *Isotope and Radiation Techniques in Soil Physics and Irrigation Studies*, International Atomic Energy Agency, Vienna, 335–340.
Gee, G. W., and Hillel, D. (1988). ''Groundwater recharge in arid regions: Review and critique of estimation methods.'' *Hydrolog. Processes*, 2, 255–266.
Gray, D. M., Granger, R. J., and Dyck, G. E. (1985). ''Overwinter soil moisture changes.'' *Trans. Am. Soc. of Agric. Engrs*., 28, 442–447.
Gray, D. M., and Granger, R. J. (1986). ''In situ measurements of moisture and salt movement in freezing soils.'' *Can. J. Earth Sci*., 23(5), 696–704.
Hendry, M. J. (1988). ''Hydrolgeology of clay till in a prairie region of Canada.'' *Groundwater*, 26(5), 607–614.
Hobbs, E. H., and Krogman, K. K. (1971). ''Overwinter precipitation storage in irrigated and nonirrigated Chin Loam soil.'' *Can. J. Soil Sci*., 51, 13–18.
Hoekstra, P. (1966). ''Moisture movement in soils under temperature gradients with cold side temperature below freezing.'' *Water Resour. Res*., 2(2), 241–250.
Hoogmoed, W. B., and Bouma, J. (1980). ''A simulation model for predicting infiltration into cracked clay soil.'' *Soil Sci. Soc. of Am. J*., 44, 458–461.
International Atomic Energy Agency. (1991). Data for Anchorage, Alaska, 〈http://www.iaea.or.at/programs/ri/gnip/gnipinfo./htm#cont_3〉.
Joshi, B. (1997). ''Estimation of diffuse vadose zone soil-water flux in a semi-arid region,'' PhD dissertation, Dept. of Agricultural and Bioresource Engineering, University of Saskatchewan, Saskatoon, Canada.
Keller, K. C., van der Kamp, G., and Cherry, J. A. (1986). ''Fracture permeability and groundwater flow in clayey till near Saskatoon Saskatchewan.'' *Can. Geotech. J*., 23, 229–240.
Li, Y., and Ghodrati, M. (1997). ''Preferential transport of solute through soil columns containing constructed macropores.'' *Soil Sci. Soc. of Am. J*., 61(5), 1308–1317.
Maulé, C. P., Black, M., and Chanasyk, D. (1993). ''Snowmelt erosivity of the Canadian Prairies.'' *Paper No. 932097*, Int. Summer Meeting, American Society of Agricultural Engineers/Canadian Soc. of Agric. Engrs., Spokane, Wash., June 20–23.
Maulé, C. P., Chanasyk, D. S., and Muehlenbachs, K. (1994). ''Isotopic determination of snow-water contribution to soil water and groundwater.'' *J. Hydro*., 155, 73–91.
Oliphant, J. L., Tice, A. R., and Nakano, Y. (1983). ''Water migration due to a temperature gradient in frozen soil.'' *Proc., 4th Int. Conf. on Permafrost*, Fairbanks, Alaska, 951–956.
Payne, B. R. (1972). ''Isotope hydrology.'' *Adv. in Hydrosci*., 8, 95–138.
Peck, R. J., Johnston, C. D., and Williamson, D. R. (1981). ''Analysis of solute distribution in deeply weathered soils.'' *Agric. Water Mgmt*., 4, 83–102.
Porter, L. K., Kemper, W. D., Jackson, R. D., and Stewart, B. A. (1960). ''Chloride diffusion in soils as influenced by moisture content.'' *Soil Sci. Soc. of Am. J*., 24, 460–463.
Rhoades, J. D. et al. (1997a). ''Salt distributions in cracking soils and salt pickup by runoff waters.'' *J. Irrig. and Drain. Engrg*., ASCE, 123(5), 323–328.
Scanlon, B. R. (1991). ''Evaluation of moisture flux from chloride data in desert soils.'' *J. Hydro*., 128, 137–156.
Seyfried, M. S., and Rao, P. S. C. (1987). ''Solute transport in undisturbed columns of an aggregated tropical soil: Preferential flow effects.'' *Soil Sci. Soc. of Am. J*., 51, 1434–1444.
Shouse, P. J. et al. (1997). ''Salt transport in cracking soils: Bromide tracer study. Salt distributions in cracking soils and salt pickup by runoff waters.'' *J. Irrig. and Drain. Engrg*., ASCE, 123(5), 329–335.
Singh, P., and Kanwar, R. S. (1991). ''Preferential solute transport through macropores in large undisturbed saturated soil columns.'' *J. Envir. Quality*, 20, 295–300.
Solomon, D. K., Poreda, R. J., Cook, P. G., and Hunt, A. (1995). ''Site characterization using $^{3}H/^{3}He$ ground water ages, Cape Cod, MA.'' *Groundwater*, 33(6), 988–996.
Souster, W. E. (1979). ''Soils of the Kernen crop research farm.'' *Publ. M51*, Saskatchewan Inst. of Pedology, University of Saskatchewan, Saskatoon, Canada.
Stephens, D. B. (1994). ''A perspective on diffuse natural recharge mechanisms in areas of low precipitation.'' *Soil Sci. Soc. of Am. J*., 58(1), 40–48.
Sudicky, E. A. (1983). ''An advection-diffusion theory of contaminant transport for stratified media,'' PhD thesis, University of Waterloo, Waterloo, Ont., Canada.
Sudicky, E. A., and McLaren, R. G. (1992). ''The Laplace Transform Galerkin technique for large scale simulation of mass transport in discretely fractured porous formations.'' *Water Resour. Res*., 28(2), 499–514.
Thomas, G. W., and Phillips, R. E. (1979). ''Consequences of water movement in macropores.'' *J. Envir. Quality*, 8(2), 149–152.
White, R. E. (1985). ''The influence of macropores on the transport of dissolved and suspended matter through soil.'' *Adv. Soil Sci*., 3, 95–113.
Wittrock, V., Begrand, R. M., and Wheaton, E. E. (1989). ''Saskatoon SRC climatological reference station summary, 1988.'' *Tech. Rep. 222*, Saskatchewan Research Council, Saskatchewan, Canada.

J. Irrig. Drain Eng. 1999.125:122-130.